

Machine Learning and Representation Basics

Controlled RAG Evaluation Corpus - Book 3

Purpose

This synthetic mini-book is designed for a beginner-friendly RAG engineering laboratory. It contains stable facts, overlapping terminology, and controlled distractors so retrieval and generation changes can be evaluated reproducibly.

Corpus rule: keep this file unchanged once the baseline evaluation dataset has been created.

Chapter 1: 1. Learning from Data

Supervised learning uses examples paired with target labels or values. Classification predicts categories, while regression predicts numeric values. Unsupervised learning looks for structure without target labels.

A training set is used to fit model parameters. A validation set helps choose configurations, and a test set estimates performance after choices have been made. Repeatedly optimizing against the test set weakens its value as an independent estimate.

Overfitting occurs when a model captures training-specific patterns that do not generalize well. Regularization, more representative data, and careful validation can reduce overfitting.

Chapter 2: 2. Representations and Embeddings

An embedding is a dense numeric representation learned or constructed so that useful relationships can be expressed geometrically. Text embedding models map passages and queries into vectors that can be compared for semantic retrieval.

Cosine similarity compares the angle between vectors, while Euclidean distance measures straight-line distance. The interpretation of a score depends on the metric and system, so a raw number should not automatically be treated as a universal probability of relevance.

Embedding quality is task dependent. A model that works well for general English may behave differently on source code, legal text, biomedical text, or multilingual documents.

Chapter 3: 3. Evaluation Principles

An evaluation metric is useful only when it measures behavior connected to the application's goal. Accuracy can be appropriate for balanced classification but misleading when classes are highly imbalanced.

Offline evaluation uses a fixed dataset to compare systems before deployment. Online evaluation observes behavior in production, where traffic and user behavior can differ from the curated offline dataset.

A baseline is a reference configuration used for comparison. Changing one important variable at a time makes causal interpretation easier than changing chunking, embeddings, retrieval, prompting, and the model simultaneously.

Chapter 4: 4. Ranking and Retrieval

Information retrieval systems rank candidate items for a query. Precision emphasizes how many retrieved items are relevant, while recall emphasizes how much of the relevant information was successfully retrieved.

A reranker takes an initial candidate set and performs a more expensive relevance assessment to improve ordering. This can increase precision but also adds latency and computational cost.

Top-k retrieval returns a fixed number of candidates. A large k may improve recall but can introduce irrelevant context; a small k can be precise yet omit necessary evidence.

Chapter 5: 5. Evaluation Distractors

A model's training loss is not the same as a RAG system's answer faithfulness. Training loss concerns model optimization, while faithfulness asks whether a generated answer is supported by provided context.

High retrieval similarity does not guarantee factual correctness. A highly similar passage can itself be outdated, misleading, poisoned, or unrelated to the exact claim being generated.

Latency is a system metric rather than a semantic-quality metric, but it matters because an improvement in answer quality may be unacceptable if it makes the application too slow or expensive.

Evaluation Notes

This book is synthetic and intentionally compact. Facts are written explicitly so they can support reference answers, retrieval-context checks, groundedness tests, citation experiments, and regression tests. Some passages are deliberate distractors and are labeled as such.